

The Bosonic Pt Particle: $m \in \mathbb{M}^D$

$$S = -m \int dt, \quad ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = -d\tau^2$$

$$\therefore dt^2 = -\eta_{\mu\nu} dx^\mu dx^\nu \Rightarrow S = -m \int \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

$$S = -m \int d\tau \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

• Point Particle extremizes its proper time

• Parameterization: $\tau \rightarrow \tau'(\sigma) = \tau - f(\tau)$

$$x^\mu(\tau') = x^\mu(\tau) \quad (\text{Scalar})$$

$$\text{Variation: } \delta\phi(\xi) = \phi'(\xi) - \phi(\xi)$$

$$\delta x^\mu = x^\mu(\tau) - x^\mu(\tau')$$

$$x^\mu(\tau) = x^\mu(\tau') + f(\tau') \frac{d}{d\tau'} x^\mu(\tau')$$

$$\therefore \delta x^\mu(\tau) = f(\tau) \dot{x}^\mu(\tau)$$

Feynman Formulation: $S = \frac{i}{2} \int d\tau \{ e^\mu \dot{x}^\mu \dot{x}^\nu \eta_{\mu\nu} - m^2 e \}$

x^μ & e are separate fields.

$$\Rightarrow e_\mu^a = \frac{\partial x^a}{\partial x^\mu} e_\mu^b \quad x^a = \tau' \quad \boxed{\delta e = f \dot{e} + \dot{f} e}$$

$$e_\mu^a(\tau') = (1 - \dot{f}) e_\mu^b$$

$$e_\mu^a(\tau) = f \dot{e}_\mu^a = (1 - \dot{f}) e_\mu^b$$

$$\delta e_\mu^a = e_\mu^a(\tau) - e_\mu^a(\tau') = f \dot{e} + \dot{f} e$$

* Scalar field Coupled to gravity

$$\int d^D x \, e \, \partial_\mu x^a \partial^\mu x_a, \quad m \in \mathbb{D} \quad \mu \rightarrow t$$

$$\int d^D x \, e \, \partial_t x^a \partial_t x_a$$

and since this is 1-D. we have

$$e_\mu^t \eta_{tt} e_\nu^t = g_{\mu\nu} \quad \text{and so the } \delta(\det(e_\mu^a)) = e_\mu^a$$

and thus $g_{\mu\nu}$ must be 1D as well

$$e_\mu^t \eta_{tt} e_\sigma^t = g_{\mu\sigma}$$

$$e_\tau^t \equiv e$$

$$-e^2 = g_{\tau\tau}$$

- Now express the action to $\mathcal{O}(\epsilon^2)$ and introduce conj. momentum $P_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

$$P_\mu = e^{-1} \dot{x}^\nu g_{\mu\nu}$$

we can solve for e using the on-shell condition $P_\mu P^\mu = -m^2$

$$(e^{-1} \dot{x}^\nu g_{\mu\nu}) (e^{-1} g^{\mu\alpha} \dot{x}^\lambda g_{\lambda\alpha}) = -m^2$$

$$e^{-2} \dot{x}^\nu g_{\mu\nu} \dot{x}^\mu = -m^2$$

- Actually not sure... but we have

$$S = \int d\tau \left\{ \dot{x}^\mu P_\mu - \frac{e}{2} (P^\mu P_\mu + m^2) \right\}$$

Recover previous form by using P_μ E.O.M.

i.e. the previous is on-shell.

$$\text{E.O.M: } \frac{d}{d\tau} \left(e^{-1} g_{\mu\nu} \dot{x}^\nu - m \dot{x}^\mu \right) = 0$$

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0$$

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial e} \right) - \frac{\partial \mathcal{L}}{\partial e} = 0$$

$$\frac{d}{d\tau} \left(-\frac{1}{2e^2} \dot{x}^2 - m^2 \right) = 0 \Rightarrow \frac{d}{d\tau} \left(\frac{\dot{x}^2}{e} \right) = 0$$

$$\text{or } \frac{d}{d\tau} (e^{-1} \dot{x}^2) = 0$$

$$e^2 m^2 + \dot{x}^2 = 0$$

and the on-shell condition. $P^2 = -m^2$

⊗: substituting this into the above we find

$$e = \frac{1}{m} \sqrt{-\dot{x}^2}$$

$$S = \frac{i}{2} \int d\tau \left(\frac{m}{\sqrt{-\dot{x}^2}} \dot{x}^2 - m^2 \left(\frac{\sqrt{-\dot{x}^2}}{m} \right) \right)$$

$$= \frac{i}{2} \int d\tau (-m \sqrt{-\dot{x}^2} - m \sqrt{-\dot{x}^2}) = -m \int d\tau \sqrt{-\dot{x}^\mu \dot{x}_\mu}$$

Which is the original action.

* Dig: Poincaré invariant,

$S_{2/3}$: Can be used for massless particles $m=0$.

* Also invariant under PTT's

for $m=0$,

* From S_1 we have

$$p^\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}_\mu} = \frac{m \dot{x}_\mu}{\sqrt{-\dot{x}^\nu \dot{x}_\nu}} = \gamma m \dot{x}_\mu$$

and so $p^2 + m^2 = 0$

Note: $\dot{x}^\mu \dot{x}_\mu = \dot{x}^i \dot{x}_i$

$\vec{v} = (\dot{x}_1, \dot{x}_2, \dot{x}_3) \Rightarrow |\vec{v}|^2 = \dot{x}^i \dot{x}_i$

$(\dot{x}^0)^2 = \left(\frac{dt}{d\tau}\right)^2$ Affine time, then $=1$

and we are left with

$p^\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}_\mu} = m \dot{x}_\mu / \sqrt{1 - v^2} = \gamma m \dot{x}_\mu \quad (c=1)$

BRST Quant Review: $[T_a, T_b] = f_{ab}^c T_c$

$\Delta_\mu = g A_\mu^a T_a \mid \text{Tr}(T_a T_b) = -C_F \delta_{ab}$

$F_{\mu\nu} = g F_{\mu\nu}^a T_a = [D_\mu, D_\nu]$

$S_{\text{YM}} = \frac{1}{2g^2 C_F} \int d^4x \text{Tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{2} \int d^4x F_{\mu\nu}^a F^{\mu\nu a}$

* Gauge fixing condition $G(A_\mu) = 0$

(i) $\partial_\mu A^\mu = 0 = G(A_\mu)$ (Gauge choice)

→ We incorporate this into the action since

$G(A_\mu) = 0$ then, of course

$S_{\text{YM}} + \int \mathcal{L}(G(A_\mu)) = S_{\text{YM}}$

We do so by introducing a (Bosonic) Lag. multiplier.

$S^{\text{gf}} = - \int d^4x \text{Tr}(B G(A_\mu)) \mid B = B^a T_a \in \mathfrak{g}^{\otimes 0}$

* The new action is given by

$S = S_{\text{YM}} + S^{\text{gf}}$

this action is not Gauge invariant under

$A_\mu \rightarrow U^\dagger A_\mu U + U^\dagger \partial_\mu U \mid U = \exp(i \omega^a(x) T_a)$

$$e^{-i \omega^a T_a} A_\mu e^{i \omega^a T_a} + e^{-i \omega^a T_a} \partial_\mu (e^{i \omega^a T_a})$$

$$= \left(\frac{1}{i} - \omega^a T_a \right) A_\mu T_b \left(\frac{1}{i} + \omega^b T_b \right) +$$

$$\left\{ + (1 - \omega^2 T_0) \partial_\mu \omega^c T_c (1 + \omega^2 T_0) \right\} \\ = \underline{A_\mu} + \partial_\mu \omega \leftarrow \text{"U(1)" Gauge trans.}$$

Gauge invariance will be replaced by the BRST symmetry.

* Take $U = \exp\{\omega^a T_a\} \mapsto \exp\{\Lambda^a T_a\}$
i.e. $\omega^a \rightarrow \Lambda^a$.

where c^a : Grassmann odd parameter (Ghost)

$$\delta A_\mu = D_\mu \omega \mapsto \delta A_\mu = \Lambda D_\mu c$$

$$\delta A_\mu = \Lambda D_\mu c \equiv \Lambda (\partial_\mu c + [A, c])$$

Ghost transformations:

$$\delta c = -\Lambda c \cdot c = -\frac{\Lambda}{2} \{c, c\} \quad \left| \begin{array}{l} c \cdot c = \text{(matrix)} \\ \text{mult.} \end{array} \right.$$

$$\delta c^e = -\frac{\Lambda}{2} c^a c^b f_{ab}^e$$

Anti-Ghost transf: $b \equiv b^a T_a$

$$\delta b = \Lambda b, \quad \delta B = 0$$

$$\delta^2_{\text{BRST}} = 0$$

* The full action:

$$S^{\text{gh}} = - \int d^4x \text{Tr} \{ b(x) \delta G(x) \}$$

$$\text{But } \delta G(x) = \int d^4y \frac{\delta G(x)}{\delta A^\mu(y)} \delta A^\mu(y)$$

$$\text{and } \delta A^\mu(x) = \Lambda D^\mu c$$

Thus:

$$S^{\text{gh}} = - \int d^4x \text{Tr} \left\{ b(x) \int d^4y \frac{\delta G(x)}{\delta A^\mu(y)} \Lambda D^\mu c \right\}$$

The full Action reads

$$S = S_{\text{gauge}} + S^{\text{gh}} + S^{\text{gf}}$$

$$S = \frac{1}{2g^2 c_F} \int d^4x \text{Tr} \{ F_{\mu\nu} F^{\mu\nu} \} - \int d^4x \text{Tr} \{ B(x) G(A_\mu) \} \\ = \int d^4x \text{Tr} \left\{ b(x) \int d^4y \frac{\delta G(x)}{\delta A^\mu(y)} \Lambda D^\mu c \right\}$$

* Lorentz Gauge we have: $G = \partial^\mu A_\mu$

$$\begin{aligned} S^{gh} &= \int d^4x \text{Tr} \{ b(x) \int d^4y \partial_\mu^\dagger \delta(x-y) \Delta D^\mu C(y) \} \\ &= + \int d^4x \text{Tr} \{ b(x) \int d^4y \delta(x-y) \partial_\mu^\dagger (\Delta D^\mu C(y)) \} \\ &= \int d^4x \text{Tr} \{ b(x) \Delta \partial_\mu D^\mu C(x) \} \end{aligned}$$

* Note: $\Delta(S^{gh} + S^f) = - \int d^4x \delta_A \text{Tr} \{ b(x) G[A^a(x)] \}$

$f(\tau) \rightarrow \Delta C(\tau)$, Δ grassmann odd
(Recall $\tau' = \tau - f(\tau)$) and

$$\delta x^\mu = f \dot{x}^\mu, \quad \delta e = f \dot{e} + \dot{f} e$$

$$\delta x^\mu = (\Delta C) \dot{x}^\mu, \quad \delta e = \frac{d}{d\tau} (\Delta C e)$$

* For ΔC to be real i.e. $(\Delta C)^\dagger = \Delta C$

we take $C^\dagger = C$, $\Delta^\dagger = -\Delta$

$$(\Delta C)^\dagger = C^\dagger \Delta^\dagger = -C \Delta = \Delta C \quad \checkmark$$

$$\delta x^\mu = (\Delta C) \dot{x}^\mu \quad \text{and} \quad \delta \dot{x}^\mu = 0$$

$$0 = \Delta \delta C \dot{x}^\mu + \Delta C \delta \dot{x}^\mu$$

$$\Delta \delta C \dot{x}^\mu + \Delta C (\Delta \dot{C} \dot{x}^\mu) + (\Delta C) (\Delta C) \dot{x}^\mu = 0$$

$$\Delta^{-1}(\cdot) = \delta C \dot{x}^\mu + C \Delta \dot{C} \dot{x}^\mu + C \Delta C \dot{x}^\mu = 0$$

$$\delta C \dot{x}^\mu + C \Delta \dot{C} \dot{x}^\mu + C^2 \Delta \dot{x}^\mu = 0$$

but $C^2 = 0$ so

$$\delta C \dot{x}^\mu = -C \Delta \dot{C} \dot{x}^\mu$$

$$\delta C = -C \Delta \dot{C} \quad \text{or}$$

$$\boxed{\delta C = (\Delta C) \dot{C}}$$

$$[\delta_1, \delta_2] x^\mu = \delta_1(f_2 \dot{x}^\mu) - \delta_2(f_1 \dot{x}^\mu)$$

$$= f_2 (\dot{f}_2 \dot{x}^\mu + f_2 \ddot{x}^\mu)$$

$$- f_1 (\dot{f}_1 \dot{x}^\mu + f_1 \ddot{x}^\mu)$$

$$- f_2 \dot{f}_2 \ddot{x}^{\mu}$$

$$= f_{12} \ddot{x}^{\mu}$$

$$\therefore f_{12} \equiv f_2 \dot{f}_1 - f_1 \dot{f}_2$$

Since $\delta C^a = \frac{1}{2} \tilde{f}_2^a$, $[\delta_1, \delta_2] = \delta_{f_{12}}$

Taking $f \rightarrow \Lambda C$

$\delta \lambda \rightarrow 0$, $\delta b = \Lambda \lambda$

*Gauge fixing terms:

$\therefore e(\tau) \approx 1$

$$\begin{cases} G = \ln(e) = 0 \\ \text{or } G(\tau) = 1 - e \end{cases}$$

• $A^{\text{gf}} = \int d\tau \lambda \ln(e)$

$$S^{\text{BRST}} = \frac{1}{2} \int d\tau \{ e^{-2} \dot{x}^{\mu} \dot{x}^{\nu} \eta_{\mu\nu} - m^2 e \} + \int d\tau \lambda \ln(e)$$

$$- \int d\tau b D_{\tau} C$$

$$D_{\tau} C \equiv \partial_{\tau} C + \frac{d(\ln(e))}{d\tau} C$$

Quantum theory: $\int D e D x^{\mu} D C D b D \lambda e^{i A^{\text{BRST}}}$

$$\lambda = 1: \int D e \dots \left(\int D \lambda e^{\int d\tau \lambda \ln(e)} \right) e^{i(S^{\text{orb.}} + S^{\text{gh}})}$$

$$= \int D e \dots \delta(\ln(e)) e^{i(S^{\text{orb.}} + S^{\text{gh}})}$$

This sets $\ln(e) = 0 \Rightarrow e = 1$ ✓

$$\int D x^{\mu} D C D b \exp \left\{ i \left(\frac{1}{2} \int d\tau \dot{x}^{\mu} \dot{x}^{\nu} \eta_{\mu\nu} - m^2 \right) - \int d\tau b D_{\tau} C \right\}$$

Further:

$$D_{\tau} e = \dot{e} + \frac{1}{e} \frac{de}{d\tau} C, \quad \frac{de}{d\tau} = 0$$

$$D_{\tau} C = \dot{C}$$

$$Z[b, c] = \int D x^{\mu} D C D b e^{i \left(\frac{1}{2} \int d\tau \dot{x}^{\mu} \dot{x}^{\nu} \eta_{\mu\nu} - m^2 - \int d\tau b \dot{C} \right)}$$

(We must replace λ in: $\delta b = \Lambda \lambda$)

Use e E.O.M

$$\partial_{\tau} \frac{\partial \mathcal{L}}{\partial (\dot{e})} - \frac{\partial \mathcal{L}}{\partial e} = 0$$

$$\delta \mathcal{L} / \delta e = -\frac{1}{2} e^{-2} \dot{x}^{\mu} \dot{x}^{\nu} \eta_{\mu\nu} - \frac{1}{2} m^2 + \lambda \frac{1}{e}$$

$$\delta \mathcal{L} / \delta \dot{e} = b e^{-1} \dot{C}$$

$$\therefore \partial_{\tau}(\lambda) = \frac{d}{d\tau}(-bc)$$

$$\therefore -\frac{d}{d\tau}(bc) - \left(-\frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} - \frac{1}{2}m^2 + \lambda\right) = 0$$

$$\therefore \lambda = \frac{d}{d\tau}(bc) + \frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} + \frac{1}{2}m^2$$

$\therefore$

$$\delta b = \Lambda \lambda = \Lambda \left(\frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} + \frac{1}{2}m^2 - \frac{d}{d\tau}(bc) \right)$$

Noether charge:

$$S^{\text{total}} = \int d\tau \left(\frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} - \frac{1}{2}m^2 - b\dot{c} \right)$$

with

$$\delta x^{\mu} = (\Lambda c)\dot{x}^{\mu}$$

$$\delta b = \Lambda \left(\frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} + \frac{m^2}{2} - \frac{d}{d\tau}(bc) \right)$$

$$\delta c = (\Lambda c)\dot{c}$$

* Use Λ as the symm. parameter:

$$\delta_{\text{total}} S^{\text{total}} = \int d\tau \left\{ \dot{x}^{\mu}(\delta x_{\mu}) - \delta b \dot{c} - b(\delta \dot{c}) \right\}$$

$$= \int d\tau \dot{x}^{\mu} \Lambda (\dot{c} \dot{x}_{\mu} + c \ddot{x}_{\mu}) - \Lambda (\dot{c} - \dots)$$

$$= \int d\tau \left(\dot{x}^{\mu} \Lambda \dot{c} \dot{x}_{\mu} + \dot{x}^{\mu} \Lambda c \ddot{x}_{\mu} - \Lambda \left(\frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} + \frac{m^2}{2} - \frac{d}{d\tau}(bc) \right) \dot{c} - b \Lambda \frac{d}{d\tau}(c \dot{c}) \right)$$

$$= \int d\tau \left(\Lambda c (\dot{x}^{\mu}\dot{x}_{\mu}) - \dot{x}^{\mu} \Lambda c \ddot{x}_{\mu} + \frac{1}{2} \Lambda \dot{x}^{\mu}\dot{x}_{\mu} + \Lambda b c + b \Lambda c \dot{c} \right)$$

$\Rightarrow$

$$= \Lambda \left(2\dot{x}^{\mu}\dot{x}_{\mu} c + \frac{1}{2}\dot{x}^{\mu}\dot{x}_{\mu} c + m^2 c \right)$$

* These factors don't match, the $2 \rightarrow \frac{1}{2}$
 ... Not sure how...

$$p^{\mu} \equiv \dot{x}^{\mu}$$

$$\therefore \boxed{T^0 = c(p^{\mu}p_{\mu} + m^2)}$$

There are no SPATIAL Dirs to int. over
 thus

$$Q = c(p^m p_m + m^2)$$

Poisson Bracket for Spm $[-1/2, 1/2]$:

Configuration Space: (q_A, p_A)

$$\{f, g\}_{P.B.} \equiv \sum_A \left\{ f \frac{\partial}{\partial q_A} \frac{\partial}{\partial p_A} g - (-1)^{fg} g \frac{\partial}{\partial q_A} \frac{\partial}{\partial p_A} f \right\}$$

$$(-1)^{fg} \equiv \begin{cases} -1 & \text{if } f \text{ \& } g \text{ are Grassmann odd} \\ +1 & \text{otherwise} \end{cases}$$

$$\{f, g\}_{P.B.} = -(-1)^{fg} \{g, f\}_{P.B.} \quad | \quad \{f, gk\}_{P.B.} = (-1)^{fg} g \{f, k\}_{P.B.} + \{f, g\}_{P.B.} k$$

$$\{f_1 + f_2, g\}_{P.B.} = \{f_1, g\}_{P.B.} + \{f_2, g\}_{P.B.}$$

Point Particle: $\{x^\mu, p^\mu\}_{P.B.} = \eta^{\mu\nu} \quad | \quad \{c, b\}_{P.B.} = 1$

$$H = p^\mu p_\mu + m^2$$

$$\delta(\cdot) = \{ \cdot, \Delta Q \}_{P.B.} \quad | \quad \{Q, Q\} = 0$$

Quantization: (Dirac Rule)

$$\{ \cdot, \cdot \}_{P.B.} \mapsto \begin{cases} \frac{i\hbar}{2} \{ \cdot, \cdot \} & \text{for two odd quantities} \\ \frac{i\hbar}{2} [\cdot, \cdot] & \text{otherwise} \end{cases}$$

$\therefore$

$$[x^\mu, \hat{p}^\nu] = i\hbar \eta^{\mu\nu} \quad | \quad \{c, b\} = i\hbar$$

In the Schrödinger rep:

$$x^\mu \rightarrow x^\mu, \quad c \rightarrow c, \quad p^\mu \rightarrow -i\hbar \frac{\partial}{\partial x^\mu}, \quad b \rightarrow i\hbar \frac{\partial}{\partial c}$$

$$Q = c(p^\mu p_\mu + m^2) \rightarrow \hat{Q}$$

$$\hat{Q} = \hat{c}(-\partial^2 + m^2)$$

NilPotency comes from the fact that $\hat{c}^2 = 0$.

$\Psi(x^\mu, c)$: wave fun

• Expand in c

$$\Psi(x^\mu, c) = \Psi(x^\mu) + c \frac{\partial \Psi(x^\mu, c)}{\partial c}$$

Physical interpretation: The first term is the wave function for the particle, and the second term is the wave function for the particle with one unit of charge.

i.e. we get rid of 'redundant counting'
 $\psi_1 \sim \psi_2 \Leftrightarrow \exists \psi_3 \text{ s.t.}$

$$\psi_1 - \psi_2 = Q\psi_3$$

- Note, for a state to be invariant under a BRST transform

$$e^{iQ}|\psi\rangle = |\psi\rangle$$

$$\text{or } e^{iQ}|\psi\rangle = |\psi\rangle + iQ|\psi\rangle$$

Thus, the physical Requirement is that

$$Q|\psi\rangle = 0$$

But $Q = c(-\partial^2 + m^2)$, so in field theory this is equivalent to

$$c(-\partial^2 + m^2)\psi = 0, \quad c \neq 0$$

$$(-\partial^2 + m^2)\psi = 0$$

The Super Point Particle:

(i) Spinning Particle: World-line susy

(ii) Brink-Schwarz Particle: Space-time susy

- Begin with a theory with D fields x^μ and their susy partners χ^μ

$$\text{Consider: } S^F = \frac{1}{2} \int d\tau (\dot{x}^\mu \dot{x}^\mu - i \dot{\chi}^\mu \chi^\mu) m_{\text{new}}$$

(i) Time translations: $\tau = \tau' + a$

$$x^\mu(\tau) = x^\mu(\tau') + a \dot{x}^\mu(\tau')$$

$$\delta x^\mu = a \dot{x}^\mu(\tau)$$

$$\delta \chi^\mu = a \dot{\chi}^\mu(\tau)$$

$\alpha_\alpha, \alpha_\beta \sim \omega_\alpha \dot{x}_\alpha$
 Rigid susy: $\delta x^m = i\epsilon \chi^m$, $\delta \chi^m = \dot{x}^m \epsilon$
 $\epsilon^* = \epsilon$, $\chi^* = \chi$

We note

$$[\delta_1, \delta_2](\cdot) = 2i\epsilon_1 \epsilon_2 \frac{d}{d\tau}(\cdot)$$

Commutator of susy $\sim$ time translation
 of magnitude $2i\epsilon_1 \epsilon_2$

$$\delta x^m = i\epsilon \chi^m \Rightarrow \delta \dot{x}^m = i\dot{\epsilon} \chi^m + i\epsilon \dot{\chi}^m$$

$$\delta \chi^m = \dot{x}^m \epsilon \Rightarrow \delta \dot{\chi}^m = \ddot{x}^m \epsilon + \dot{x}^m \dot{\epsilon}$$

$$\delta S = \int d\tau \left(i\epsilon \chi^m \dot{\chi}^m \eta_{mn} - i\dot{x}^m \epsilon \dot{\chi}^m \eta_{mn} \frac{1}{2} \right. \\ \left. + \frac{1}{2} i\epsilon \chi^m \dot{\chi}^m \epsilon + \frac{1}{2} i\epsilon \chi^m \dot{\epsilon} \right)$$

$$\int \delta S = \dot{x}^m \chi^m \eta_{mn}$$

If we now couple the theory to the worldline
 Supergravity $(e, \psi) =$

$$S = \frac{1}{2} \int d\tau (e^{-1} \dot{x}^m \dot{x}^n - i\chi^m \dot{\chi}^n - R e^{-1} i\psi \chi^m \dot{\chi}^n) \eta_{mn}$$

* This means we must introduce local susy terms:

$$\delta x^m = i\epsilon \chi^m \quad \delta \chi^m = (\dot{x}^m - \frac{R}{2} i\psi \chi^m) \epsilon e^{-1}$$

$$\delta e = i\kappa \epsilon \psi \quad \delta \psi = \frac{2}{\kappa} \dot{\epsilon}$$

→ Invar. under rep. symm.

$$\delta\psi = \frac{d}{d\tau}(K\psi), \quad h = h(\tau)$$

Proof: (Noether Method)

start w/ $S^F = \frac{1}{2} \int d\tau (\dot{x}^\mu \dot{x}^\nu - i \dot{x}^\mu \dot{\chi}^\nu) \eta_{\mu\nu}$

$$\delta x^\mu = i\epsilon \chi^\mu \mid \delta \chi^\mu = \dot{x}^\mu \epsilon$$

- We introduce the supergravity fields (h, ψ)

which have the susy trans

$$\delta h = i\epsilon \psi \mid \delta \psi = 0$$

$$S_{\text{susy}} = \frac{1}{2\kappa^2} \int d\tau (h R - \psi_\mu \gamma^{\mu\nu\rho} D_\nu \psi_\rho)$$

with susy

$$\delta h = i\epsilon \psi, \quad \delta \psi = 0$$

- The fields also have a local abelian transf.

$$\delta h = g \mid \delta \psi = \bar{Q}$$

$$g_i \in C^\infty(M), \quad \bar{Q} \in C^\infty(M)$$

* We now let ϵ become local: $\epsilon = \epsilon(\tau)$

S^F is no longer susy invariant.

$\delta_{\text{susy}} S^F \neq 0$. Instead, we find

$$S^2 \equiv S^F - i \int d\tau \frac{\kappa}{2} \psi \dot{\psi}$$

IS inv. at order κ^0 , $\bar{Q} \equiv \frac{2}{\kappa} \epsilon$

→ Link local invariance to local susy

Thus,

$$S^2 = S^F - i \int d\tau \frac{\kappa}{2} \psi \dot{\psi}$$

Note the linearized gravity

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$$

But $g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$

$$\therefore -e^2 = -1 + \kappa h$$

$$e^2 = 1 - \kappa h$$

$$2e\delta e = -\kappa\delta h$$

$\Rightarrow$ Come Back later... Not sure about how he's introducing a linearised grav. term.

The Coupled action in the first order formalism:

$$\int d\tau \left(\dot{x}^\mu p_\mu - \frac{1}{2} e p^\mu p_\mu - \frac{i}{2} \chi_\mu \dot{\chi}^\mu - \frac{i\kappa}{2} \psi \chi^\mu p_\mu \right)$$

Susy:

$$\begin{aligned} \delta x^\mu &= i\epsilon \chi^\mu & \delta \chi^\mu &= p^\mu \epsilon, \quad \delta p^\mu = 0 \\ \delta e &= i\kappa \epsilon \psi & \delta \psi &= \frac{2}{\kappa} \epsilon \end{aligned}$$

The E.O.M are given by

$$\underline{p^\mu}: \quad \frac{\partial}{\partial \tau} \frac{\partial \mathcal{L}}{\partial p^\mu} - \frac{\partial \mathcal{L}}{\partial p_\mu} = 0$$

$\underline{\chi^\mu}: \quad$

$$\dot{\chi}^\mu - e p^\mu - \frac{i\kappa}{2} \psi \chi^\mu = 0$$

$$\dot{p}^\mu = 0$$

$\underline{e}: \quad$

$$p^\mu p_\mu = 0$$

$\underline{\psi}: \quad$

$$\chi^\mu p_\mu = 0$$

Both are Constraints

$\equiv \phi$

$\equiv \varphi$

$$\underline{\chi^\mu}: \quad -\frac{i}{2} \dot{\chi}^\mu - \frac{i}{2} \dot{\chi}^\mu + \frac{i\kappa}{2} \psi p^\mu = 0$$

$$2\dot{\chi}^\mu + \kappa \psi p^\mu = 0$$

* If we eliminate p^μ via its E.O.M then we recover the original action/transt.'s

Gauge choices: $e=1, \psi=0$

$$\dot{\phi}^\mu = 0, \quad \dot{\chi}^\mu = 0, \quad p^\mu = \dot{x}^\mu$$

Hamiltonian approach:

$$\{x^\mu, p^\nu\} = \eta^{\mu\nu} \quad \& \quad \{\chi^\mu, \chi^\nu\} = i\eta^{\mu\nu}$$

$$H = \frac{1}{2} e p_\mu p^\mu + i \frac{k}{2} \psi p_\mu \chi^\mu = a\phi + b\phi$$

$$\frac{d\phi}{d\tau} = \{\phi, H\}$$

where:

$$\{\phi, \psi\}_{\text{P.B.}} = 0 \quad \& \quad \{\phi, \phi\}_p = i\phi$$

Dirac Method: Momentum conj. to χ^μ is $\frac{i}{2}\chi^\mu$

with $p_\chi^\mu = \frac{i}{2}\chi^\mu$ we must adopt the constraint

$$S^\mu \equiv p_\chi^\mu - \left(\frac{i}{2}\right)\chi^\mu = 0$$

Starting from

$$\{\chi^\mu, p_\chi^\nu\} = \eta^{\mu\nu}$$

we discover $S^\mu = 0$ is second class so the

$\{\phi, S^\mu\} \neq 0$ (another constraint)

* We instead use the Dirac Bracket

$$[x^\mu, p^\nu] = i\eta^{\mu\nu}, \quad \{\chi^\mu, \chi^\nu\} = -i\eta^{\mu\nu}$$

$$x^\mu \rightarrow \tilde{x}^\mu$$

$$p^\mu \rightarrow -i \frac{\delta}{\delta x^\mu}$$

$$\underline{\underline{t_1 = 1}}$$

where we now adopt the following

$$\chi^m = \gamma^0 / \sqrt{2}$$

$$\{ \gamma^m, \gamma^n \} = 2 \eta^{mn}$$

Constraint S: $\phi = p_m p^m = \frac{\partial}{\partial x^m} \frac{\partial}{\partial x_m} = -\partial^2$

$$\phi = p_m \chi^m = (-i \frac{\partial}{\partial x^m}) (i \gamma^m / \sqrt{2}) \quad \phi = -\partial^2$$

$$\therefore \phi = \partial / \sqrt{2}$$

* Impose: $\partial \gamma = 0$ (γ : wave fⁿ)

→ This is a massless spin-(1/2) particle

* $N=1$ worldline susy describes a spin 1/2 particle upon quantization.

BRST Quant: Start from

$$S = \frac{1}{2} \int d\tau (e^{-1} \dot{x}^\mu \dot{x}^\nu - i \chi^\mu \dot{\chi}^\nu - \kappa e^{-1} i \psi \chi^\mu \dot{x}^\nu) \mathcal{M}_{\mu\nu}$$

* For ~~EACH~~ local symmetry a ghost is introduced

- c and γ for reparametrisations and local susy respectively.

γ : Grassmann even since ϵ is odd

$$\epsilon \rightarrow \Delta \gamma$$

$$\gamma^* = -\gamma$$

* For rep: $\tau' = \tau - K(\tau)$

$$K(\tau) \rightarrow \Delta c$$

Thus, we write out the full variations:

$$\delta x^\mu = i \epsilon \chi^\mu + \kappa \dot{x}^\mu = i \Delta \gamma \chi^\mu + \Delta c \dot{x}^\mu$$

For calcul:

$$\begin{aligned} \delta_B \chi^m &= i \Delta \gamma \chi^m + \Delta c \dot{\chi}^m & \delta_B \chi^m &= \left(\dot{\chi}^m - \frac{i\kappa}{2} \gamma \chi^m \right) \Delta \gamma \dot{c} \\ \delta_B e &= i\kappa \Delta \gamma \psi + \frac{d}{d\tau} (e \Delta c) & &+ \Delta c \dot{\chi}^m \\ \delta_B \psi &= \frac{2}{\kappa} \frac{d}{d\tau} (\Delta \gamma) + \frac{d}{d\tau} (\Delta c \psi) \end{aligned}$$

* c : Reparametrizations
 γ : Super symmetry

• Ghost transformations:

$$\begin{aligned} \delta_B c &= \Delta c \dot{c} - i \Delta \gamma^2 e^{-1} \\ \delta_B \gamma &= \frac{i\kappa}{2} \Delta \gamma^2 \psi e^{-1} + \Delta c \dot{\gamma} \end{aligned}$$

* Idea: Take the full symmetry of the theory and combine into a BRST symm.

→ Introduce Anti-ghost fields and β

b : Grassmann odd

β : Grassmann even

$$\delta b = \Delta \lambda$$

$$\delta \beta = \Delta \rho$$

• Introduce the Lagrange multipliers for each of the variables

$$b \rightarrow \lambda$$

$$\beta \rightarrow \rho$$

$$\delta \lambda = 0$$

$$\delta \rho = 0$$

• Gauge fixing term $(\lambda \neq 1, \rho \neq 0)$

since $e=1$ as well as $\frac{\kappa}{2} \int d\tau \psi \rho$ since $\psi=0$

$$\therefore S^{\text{gf}} = \int d\tau \left(\lambda b e^1 + \frac{\kappa}{2} \rho \psi \right)$$

$$e=1, \quad \psi=0$$

• We must then add the anti-ghost field Action:

(i) Take $\lambda \rightarrow b$, $\rho \rightarrow \beta$ in S^{gf}

(ii) Take S^{gf} and add $\delta_B S^{\text{gf}}$

$$\int d\tau \delta_B \left(b b e^1 - \frac{\kappa}{2} \beta \psi \right) \\ = \Delta (S^{\text{gf}} + S^{\text{gh}})$$

$$S^{\text{BRST}} = S^{\text{m}} + S^{\text{gh}} + S^{\text{gf}}$$

eliminating $\lambda \dot{\varphi}$ enforces the gauge cond's.

$$S^{\text{tot}} = \int d\tau \left\{ \frac{1}{2} \dot{\chi}^{\mu} \dot{\chi}^{\mu} - i \chi^{\mu} \dot{\chi}^{\mu} \mathcal{H}_{\text{can}} - b \dot{c} - \beta \dot{\psi} \right\}$$

$$\{c, b\} = 1, \quad \{\chi, \beta\} = 1$$

$$Q_B = c p_A p^A + \gamma p_A \chi^A + i \frac{b}{2} \gamma^2 \quad [\text{charge}]$$

BRST charge is nilpotent

$$Q^2 = 0$$

Which is the BRST generator:

$$S \bullet = \{ \bullet, \Delta Q \}$$

like want: $\{ \alpha, \beta \} = i\hbar$, $\{ \alpha, \beta \} = i\hbar$
 $b = i\partial/\partial c$, $\beta = -i\partial/\partial \gamma$

• Quantum BRST charge:
 $Q = -c\partial^2 - (i/\hbar)\gamma\partial - \frac{1}{2}\gamma^2\frac{\partial}{\partial c}$

* Assume $\Phi \equiv \psi + \gamma\phi - \sqrt{2} i\partial\phi c$

$Q\Phi = -i\frac{1}{2}\partial\psi\gamma - \partial^2\psi c$

and so we adopt

$Q\Phi \rightarrow [\text{Physical states}]$

The Brink-Schwarz Superparticle:

* Spacetime supersym.

(x^μ, Θ^A) , $\mu=0, \dots, D-1$

Θ^A : Spacetime Majorana spinor

$\bar{\Theta}^A \equiv \Theta^T C$, $A=1, \dots, 2^{D/2}$ $\leftarrow$ dim of cliff. alg.

$\delta x^\mu = i\bar{\epsilon}\gamma^\mu\Theta$ / $\delta\Theta_A = \epsilon_A$

ϵ_A : constant spacetime Maj. spinor

(x^μ, Θ^A) : Superspace coordinates

$[\delta_1, \delta_2]_{\text{susy}} = (\text{spacetime translations})$

* For two Maj. spinors: $\bar{\lambda}\gamma^\mu\rho = -\bar{\rho}\gamma^\mu\lambda$

$$\Pi^M \equiv \dot{X}^M - i \bar{\Theta} \gamma^M \dot{\Theta} \quad (\text{Susy invariant})$$

proof:

$$\begin{aligned} \delta_{\text{Susy}} \Pi &= (\delta \dot{X}^M) - i \delta \bar{\Theta} \gamma^M \dot{\Theta} - i \bar{\Theta} \gamma^M \delta \dot{\Theta} \\ &= i \bar{E} \gamma^M \dot{\Theta} - i \bar{E} \gamma^M \dot{\Theta} - i \bar{\Theta} \gamma^M \dot{\Theta} \\ &= 0 \end{aligned}$$

* Under a Lorentz trans. of $Sp(1, D-1)$
 $\delta_L X^M = \omega_{\nu}^M X^\nu$, $\delta_L \Theta = \frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \Theta$

Lorentz:

$$\begin{aligned} \therefore \delta_L \Pi^M &= \omega_{\nu}^M \dot{X}^\nu - i \left(\frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \Theta \right)^T C \gamma^M \dot{\Theta} \\ &= i \bar{\Theta} \gamma^M \left(\frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \Theta \right) \\ &= \omega_{\nu}^M \dot{X}^\nu - i \frac{1}{4} \omega_{\mu\nu} \Theta^T (\gamma^{\mu\nu})^T C \gamma^M \dot{\Theta} \\ &\quad - i \frac{1}{4} \omega_{\mu\nu} \bar{\Theta} \gamma^{\mu\nu} \dot{\Theta} \\ &= \omega_{\nu}^M \dot{X}^\nu \end{aligned}$$

So Π^M transforms as a vector

* We can therefore realise the following
Superspace replacement:

$$\dot{X}^M \mapsto \Pi^M$$

$$\therefore S = \frac{1}{2} \int d\tau e^{-1} \Pi^M \Pi^N \eta_{MN}$$

e is inert under susy

E.O.M: (X^M, Θ, e)

X^M :

$$\begin{aligned} \delta S &= \frac{1}{2} \int d\tau e^{-1} (\dot{X}^M - \bar{\Theta} \gamma^M \dot{\Theta}) (\dot{X}^N - \bar{\Theta} \gamma^N \dot{\Theta}) \eta_{MN} \\ \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} \right) - \frac{\delta \mathcal{L}}{\delta X^\mu} &= 0 \end{aligned}$$

$$\underline{\Theta}: \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \Theta} \right) - \frac{\partial \mathcal{L}}{\partial \Theta} = 0$$

$$\partial_\tau (e^{-1} \bar{\Theta} \gamma^\mu \pi_\mu) + e^{-1} \gamma^\mu \bar{\Theta} \pi_\mu = 0$$

• First term vanishes since

$$= \partial_\tau (e^{-1} \pi^\mu) \bar{\Theta} \gamma_\mu + e^{-1} \partial_\tau (\bar{\Theta} \gamma^\mu) \pi_\mu$$

$$= \partial_\tau (\bar{\Theta} \gamma^\mu) e^{-1} \pi_\mu = \bar{\Theta}^\tau (\gamma^\mu e^{-1} \pi_\mu)$$

$$\therefore \boxed{\gamma^\mu \pi_\mu \bar{\Theta} = 0 = \not{\pi} \bar{\Theta}} \quad (\Theta \text{ E.O.M.})$$

e:

$$-1 e^{-2} \pi^\mu \pi^\nu \eta_{\mu\nu} = 0$$

$$\therefore \boxed{\pi^\mu \pi^\nu \eta_{\mu\nu} = 0}$$

Thus: E.O.M.:

$$\boxed{\frac{\partial}{\partial \tau} (e^{-1} \pi^\mu) = 0, \not{\pi} \bar{\Theta} = 0}$$

$$\pi^\mu \pi^\nu \eta_{\mu\nu} = 0$$

$\Rightarrow$ We note that $\delta \bar{\Theta} \gamma^\mu \bar{\Theta} = -\bar{\Theta} \gamma^\mu \delta \bar{\Theta}$

* The corresponding first order action:

$$\boxed{S = \int d\tau \left\{ \pi^\mu \dot{x}^\mu - \frac{1}{2} e \dot{x}^\mu \dot{x}^\mu \right\} \eta_{\mu\nu}}$$



w/ Susy: $\boxed{\delta e = \delta p^\mu = 0}$

* $\not{\pi}$ is not invertible since

$$\not{\pi} \not{\pi} = \pi^\mu \pi_\mu = 0$$

Thus we cannot conclude $\bar{\Theta} = 0$

•  possesses an extra symmetry:

$$\delta x^\mu = i \bar{\theta} \gamma^\mu i \not{\partial} K, \quad \delta \theta = i \not{\partial} K$$

$$\delta e = -4 e \dot{\theta} K$$

K_A : is a Majorana spinor. $K_A = K_A(t)$

Claim: $\delta_\lambda S = 0$

Proof: we first show $\delta \pi^\mu = -2 \dot{\theta} \not{\partial} K$ (i)
and $\delta(\pi^\mu \bar{\psi}_\mu) = -4 \pi^\mu \bar{\psi}_\mu \dot{\theta} K$ (ii)

$$(i) \quad \delta \pi^\mu = \delta \dot{x}^\mu - i \delta \bar{\theta} \gamma^\mu \dot{\theta} - i \bar{\theta} \gamma^\mu \delta \dot{\theta}$$

$$= \frac{d}{dt} (i \bar{\theta} \gamma^\mu \delta \theta) - i \delta \bar{\theta} \gamma^\mu \dot{\theta} - i \bar{\theta} \gamma^\mu \delta \dot{\theta}$$

(ii) $\rightarrow$ trivial

- We note that the first order formalism $\int (\pi^\mu \dot{x}_\mu - \frac{1}{2} e p^\mu \dot{x}_\mu) dt$ is also k -inv. if we also impose δx^μ and e : E.O.M but take $\delta \theta = i \not{\partial} K, \delta p^\mu = 0$

* The k -symm forms a closed alg. if we are on-shell and impose $\delta x^\mu = i \bar{\theta} \gamma^\mu \delta \theta, \delta \theta = k(t) \dot{\theta}, \delta e = 0$

$\Rightarrow$ There is NO Lorentz covariant quant. for the Brink-Schwarz particle

* Conjugate Momenta of θ : λ^A

ψ

$\lambda^A = \partial \theta^A$

- first note that $P_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

$$\therefore e^{-1} \pi_\mu = P_\mu$$

$$\begin{aligned} \delta \mathcal{L} / \delta \theta^A &= i (\bar{\theta} \gamma^\mu \partial_B^A) P_\mu \\ &= i (\bar{\theta} \gamma^\mu)^A P_\mu \quad \text{or} \\ &= i (\bar{\theta} P)^A \end{aligned}$$

$\therefore$

$$\lambda^A = i (\bar{\theta} P)^A$$

Since $\frac{\partial}{\partial \theta} (\bar{\theta} \gamma^\mu \theta) = -\bar{\theta} \gamma^\mu \frac{\partial \theta}{\partial \theta}$

* This tells us that λ^A and θ^A are related, thus we can view this as an extra constant of the system

$$\bar{\chi}^A = \bar{\lambda}^A - i (\bar{\theta} P)^A = 0$$

- Factor of $\not{P}$ causes problems as it is not invertible

$$\begin{aligned} \{\theta_A, \bar{\lambda}^B\}_P &= \delta_A^B \quad \parallel \quad \{\bar{\chi}^A, \bar{\chi}^B\} = -2i (C \not{P})^A B \\ \{\theta_A, \bar{\chi}^B\} &= -2i (\not{P})_A^B \end{aligned}$$

Super Space Formulation:

(τ, θ) : Superspace of pt particle w/ worldline susp.

- Supercharges were given by

$$j = Q = \dot{x}^\mu \chi^\nu \eta_{\mu\nu}$$

In Particular, $[\delta_1, \delta_2] = [e_1 Q, e_2 Q]$ closes to

$$[\delta_1, \delta_2] = -\epsilon_1 \epsilon_2 \{Q, Q\} = 2i \epsilon_2 \epsilon_1 P$$

$$P \equiv \frac{d}{d\tau}$$

Thus, we have the following supersymmetry alg.

$$\{Q, Q\} = +2iP, \quad [Q, P] = 0$$

Since Q and P are Generators we can consider the supergroup generated by these:

$$g(\theta, \tau) = \exp(\theta Q + \tau P)$$

* Acting w/ $g(a, \epsilon) = e^{-\epsilon Q - aP}$, the pt
 $(\tau, \theta) \mapsto (\tau', \theta')$

$$\begin{aligned} g(a, \epsilon) g(\theta, \tau) &= (1 - \epsilon Q - aP)(1 + \theta Q + \tau P) \\ &= (1 - \epsilon Q - aP + \theta Q + \tau P - \epsilon Q \theta Q - a\tau P^2 \\ &\quad - aP \theta Q - \epsilon Q \tau P) \end{aligned}$$

$$= 1 + (\tau - a)P + (\theta - \epsilon)Q - a\tau P^2 - aP\theta Q - \epsilon Q\tau P$$

OR to first order:

$$e^x e^y = e^{x+y + \frac{1}{2}[x, y] + \mathcal{O}(x^2, y^2)}$$

$$\begin{aligned} \textcircled{a} &= [-\epsilon Q - aP, \theta Q + \tau P] \\ &= -[\epsilon Q, \theta Q] \end{aligned}$$

all other commutators vanish.

$$\begin{aligned} -[\epsilon^\alpha Q_\alpha, \theta^\beta Q_\beta] &= -(\epsilon^\alpha Q_\alpha \theta^\beta Q_\beta - \theta^\beta Q_\beta \epsilon^\alpha Q_\alpha) \\ &= -\epsilon^\alpha \theta^\beta Q_\alpha Q_\beta - \epsilon^\alpha \theta^\beta Q_\beta Q_\alpha \\ &= \epsilon^\alpha \theta^\beta \{Q_\alpha, Q_\beta\} \end{aligned}$$

$$\textcircled{a} = \frac{i}{2} \epsilon^\alpha \theta^\beta \{Q_\alpha, Q_\beta\} = i \epsilon^\alpha \theta^\beta P$$

$$g(a, e) g(\tau, \theta) = g((\tau - a)P + (\theta - e)Q + i e \theta P)$$

Which means that such a group action will move us along SuperSpace:

$$\tau' = \tau - a + i e \theta$$

$$\theta' = \theta - e$$

* Def: A Superfield is a field which is dependent on (τ, θ) . The field has a "Scalar" nature to it. Regly

$$\Phi'(\tau', \theta') = \Phi(\tau, \theta)$$

It is easy to generalize, as with $\mathbb{R}$ -scalar fields

$$\delta \Phi = \Phi'(\tau, \theta) - \Phi(\tau, \theta) = (aP + eQ)\Phi$$

$$g(a, e)\Phi = \Phi - \delta \Phi$$

$$\begin{aligned} \Phi'(\tau', \theta') &= \Phi'(\tau', \theta) - e \frac{d}{d\theta} \Phi'(\tau', \theta) \\ &= \Phi'(\tau, \theta) + (-a + i e \theta) \frac{d}{dt} \Phi(\tau, \theta) - e \partial_\theta \\ &= \Phi'(\tau, \theta) + (-a + i e \theta) \partial_t \Phi \end{aligned}$$

$$\begin{aligned} &- e \frac{d}{d\theta} \left[\Phi'(\tau, \theta) + (-a + i e \theta) \frac{d}{dt} \Phi(\tau, \theta) \right] \\ &= \Phi'(\tau, \theta) + (-a + i e \theta) \partial_t \Phi - e \partial_\theta \Phi + e^2 \partial_\theta^2 \Phi \\ &= \Phi'(\tau, \theta) - a \partial_t \Phi + i e \theta \partial_t \Phi - e \partial_\theta \Phi \\ &= \Phi(\tau, \theta) \end{aligned}$$

$$\begin{aligned} \therefore \delta \Phi &= \Phi'(\tau, \theta) - \Phi(\tau, \theta) \\ &= a \partial_t \Phi + e \left(\partial_\theta - i \frac{d}{dt} \theta \right) \Phi \end{aligned}$$

We conclude

$$\delta = a \frac{d}{dt} + e \left(\partial_\theta - i \frac{d}{dt} \theta \right)$$

* We define the covariant derivative:

$$D \equiv \partial_\theta + i d_\tau \theta$$

s.t.

$$\{D, Q\} = 0, \quad [d_\tau, Q] = 0$$

$$\{D, D\} = 2i d_\tau$$

We now take a scalar superfield $X(\tau, \theta)$

We know $f(\theta) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \theta^n$

But since $\theta^2 = 0$

$$f(\theta) = f(0) + \theta f^{(1)}(0)$$

Now

$$X(\tau, \theta) = X(\tau, 0) + \left(\frac{d}{d\theta} X(\tau, \theta) \right) \Big|_{\theta=0} \theta$$

Since X is a scalar field then

$X(\tau, 0)$ must be Grassmann even

while $\left(\frac{d}{d\theta} X \right)$ is odd.

$$\Rightarrow X(\tau, 0) \equiv X(\tau)$$

$$\frac{d}{d\theta} X(\tau, \theta) \Big|_{\theta=0} \equiv X'(\tau)$$

Therefore:

$$X(\tau, \theta) = X(\tau) + i\theta X'(\tau)$$

* Note $(\delta \cdot = \epsilon Q \cdot = \epsilon D \cdot) \Big|_{\theta=0}$

ex:

$$\delta X = \epsilon Q X \Big|_{\theta=0} = \epsilon D X \Big|_{\theta=0} = \epsilon i X'(\tau)$$

as well as:

$$\delta X = \delta(-i D X \Big|_{\theta=0}) = -i \epsilon Q D X \Big|_{\theta=0}$$

$$\epsilon \theta \theta \Rightarrow Q^2 = -D$$

$$\delta X = -i \epsilon D X|_{\theta=0} = -\frac{i}{2} \{D, D\} X|_{\theta=0} \\ = -\frac{i}{2} i \frac{d}{d\tau} X|_{\theta=0} = \epsilon \dot{X}$$

$\therefore$

$$\delta X(\tau) = i \epsilon \dot{X}(\tau)$$

$$\delta \dot{X}(\tau) = \dot{X} \epsilon$$

Note $\int d\theta X = i \dot{X}(\tau)$ as well as

$$\int d\tau d\theta = \int d\tau \frac{d}{d\theta} = \int d\tau D$$

if acting on any superfield

• For any scalar superfield Φ

$\int d\tau d\theta \Phi$ is susy inv.

• The original Action S for worldline susy is given by $(-i/2) \int d\tau d\theta \dot{X} D X$

Proof of invariance:

$$\delta \int d\tau d\theta X = \int d\tau d\theta (a \overset{a}{P} + \epsilon \overset{b}{Q}) X$$

We assume each field has compact support so that

$$\overset{a}{a} \int d\tau \frac{d}{d\tau} (a X) = a X|_{\partial\tau} = 0$$

$$\overset{b}{b} \int d\tau d\theta \epsilon (\partial_\theta - i \partial_\tau \theta) (X|_{\theta=0} + i \theta \dot{X}(\tau)) \\ = \underbrace{\int d\tau d\theta i \dot{X}(\tau)}_{\text{No } \theta \rightarrow 0} - \underbrace{\int d\tau d\theta i \theta \frac{d}{d\tau} X(\tau)}_{\text{compact support}}$$

Proof of Expression:

$$-\frac{i}{2} \int d\tau d\theta \dot{X} D X = -\frac{i}{2} \int d\tau D (\dot{X} D X) \\ = -\frac{i}{2} \int d\tau [D \dot{X} D X + \dot{X} D^2 X]$$

$$= i\dot{\chi} + i\theta\ddot{\chi}$$

$$(\text{again, } \theta^2 = 0)$$

$$D\dot{\chi} = i\ddot{\chi} + i\theta\ddot{\chi}$$

$$D(D\chi) = (\partial_\theta + i\frac{d}{d\tau}\theta)(i\dot{\chi} + i\theta\ddot{\chi}) \\ = i\ddot{\chi} - \theta\dot{\chi}$$

$\therefore$

$$D\dot{\chi}D\chi = (i\ddot{\chi} + i\theta\ddot{\chi})(i\dot{\chi} + i\theta\ddot{\chi}) \\ = -\dot{\chi}\dot{\chi} - \dot{\chi}\theta\ddot{\chi} - \theta\ddot{\chi}\dot{\chi}$$

$$\dot{\chi}D^2\chi = (\dot{\chi} + i\theta\ddot{\chi})(i\ddot{\chi} - \theta\dot{\chi}) \\ = i\ddot{\chi}\ddot{\chi} - \theta\ddot{\chi}\dot{\chi} - \theta\dot{\chi}\ddot{\chi}$$

Int. by parts

$$-\frac{i}{2}\int d\tau (i\ddot{\chi}\ddot{\chi} + \dot{\chi}\dot{\chi}) = \frac{2}{2}\int d\tau (\dot{\chi}\ddot{\chi} - i\dot{\chi}\dot{\chi}) \checkmark$$

Supergravity Superfield:

$$E = e(1 + i\theta\psi)$$

Coupling the scalar field χ to (super)gravity

$$(-i/2)\int d\tau d\theta E^{-1}\dot{\chi}D\chi = S_E$$

Provided we make the rescaling

$$\psi \rightarrow e^{i/2}\psi, \quad \chi \rightarrow e^{i/2}\chi$$

$$E = \det(E_a^\mu)$$

* Claim: S_E is Superconformally invariant

